

How to Get the Original Langmuir 1918 Paper

The Paper

Title: “The Adsorption of Gases on Plane Surfaces of Glass, Mica and Platinum”

Author: Irving Langmuir

Year: 1918

Journal: Journal of the American Chemical Society, Vol. 40(9), pp. 1361–1403

DOI: 10.1021/ja02242a004

Status: **PUBLIC DOMAIN** — copyright has expired

Fastest Way (2 clicks)

Option 1: Google Scholar (Recommended - Easiest)

1. Go to <https://scholar.google.com>
 2. Search: Langmuir adsorption gases 1918
 3. Look for: “The adsorption of gases on plane surfaces...”
 4. Click [**PDF**] link on the right side
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1. Go to: <https://pubs.acs.org/doi/10.1021/ja02242a004>
 2. Scroll down
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 4. PDF opens (may require clicking through one page)
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1. Go to: <https://archive.org/search.php?query=langmuir+adsorption+1918>
 2. Find the paper in results
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Citation

Langmuir, I. (1918). The adsorption of gases on plane surfaces of glass, mica and platinum. Journal of the American Chemical Society, 40(9), 1361-1403.
DOI: 10.1021/ja02242a004

Key Content from the Paper

The paper introduces the **Langmuir Isotherm equation** used throughout your project:

$$q_e = \frac{q_{max} \cdot K_L \cdot C_e}{1 + K_L \cdot C_e}$$

Where: - q_e = amount adsorbed at equilibrium - q_{max} = monolayer saturation capacity - K_L = Langmuir adsorption constant - C_e = equilibrium concentration

This foundational work is the basis for your entire Phase 2 model.

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